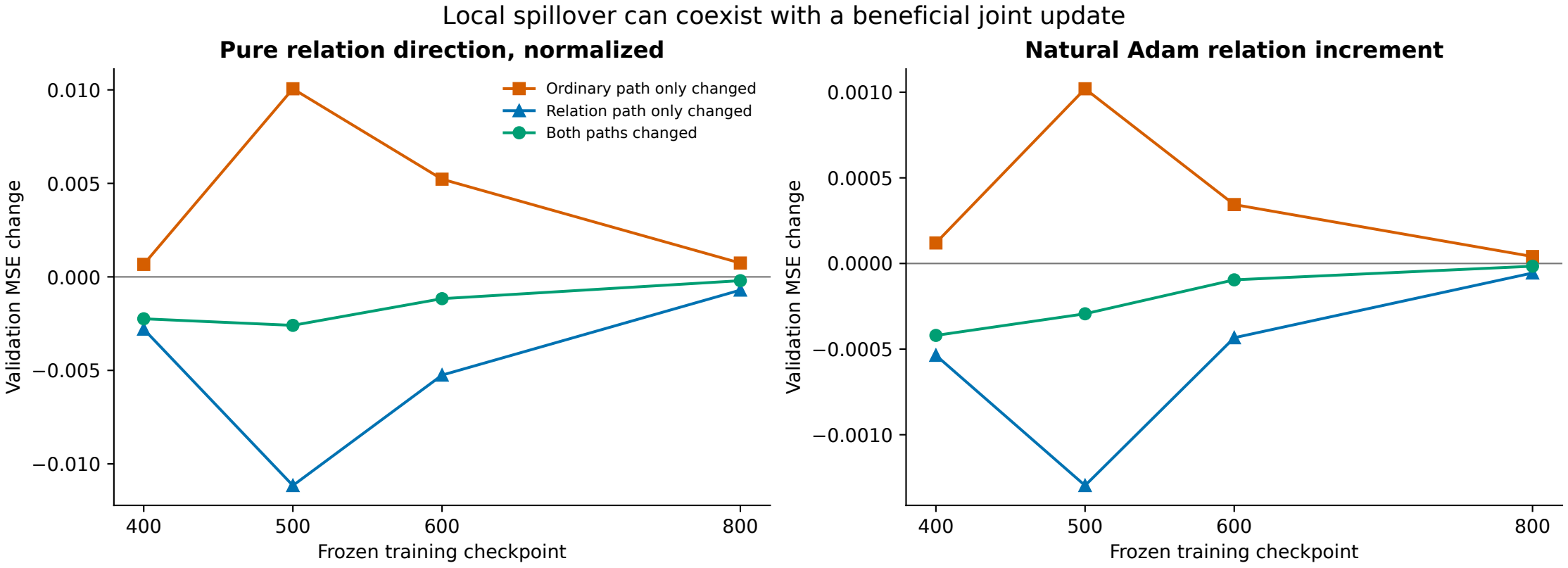
Local spillover can coexist with a beneficial joint update
Pure relation direction, normalized
Natural Adam relation increment
Ordinary path only changed
Relation path only changed
Both paths changed
Validation MSE change
0.010
0.005
0.000
−0.005
−0.010
0.0010
0.0005
0.0000
−0.0005
−0.0010
400
500
600
800
Frozen training checkpoint